

Passive Components

- In an optical network, **Passive Components** are devices that **do not require electrical power to operate on the optical signal.**
- They function using the physical properties of light (**reflection, refraction, interference, and absorption**) to direct, filter, or modify the signal.

Explanation of Passive Components

• 1. Mux and Demux (Multiplexer and Demultiplexer)

- **Explanation:** A **Mux** combines multiple light signals of different wavelengths into a single fiber.
- A **Demux** does the opposite, splitting the combined signal back into individual wavelengths at the receiving end.
- **Example:** Like a highway on-ramp where cars from different streets merge into one large highway, and an off-ramp where they split off to their specific destinations.

Explanation of Passive Components

• 2. Optical Switches

- **Explanation:** These route an optical signal from one input fiber to one of several output fibers without converting the light to electricity.
- **Example:** A railroad switch that moves the tracks to guide a train onto a different path.

• 3. Wavelength Converter

- **Explanation:** A device that changes the wavelength of an incoming signal to a different wavelength for the outgoing signal to avoid "wavelength contention" (two signals trying to use the same "lane").
- **Example:** Painting a car a different color so it can enter a lane reserved for that specific color.

Explanation of Passive Components

• 4. Couplers

- **Explanation:** These split a single optical signal into two or more fibers, or combine multiple signals into one.
- A common type is a 3dB coupler, which splits power exactly 50/50.
- **Example:** A Y-shaped water pipe that splits one stream into two.

• 5. Circulators and Isolators

- **Explanation:**
- **Isolator:** Allows light to travel in only one direction, preventing reflections from damaging the laser.
- **Circulator:** A multi-port device where light entering Port 1 exits Port 2, and light entering Port 2 exits Port 3.
- **Example:** An Isolator is like a one-way street. A Circulator is like a traffic roundabout.

Explanation of Passive Components

• 6. Attenuator

- **Explanation:** A device used to **reduce the power level** of an optical signal if it **is too strong for the receiver** (which could cause distortion or damage).
- **Example:** Wearing sunglasses to reduce the intensity of bright sunlight so you can see clearly.

• 7. Optical Filters

- **Explanation:** These **allow only a specific range of wavelengths to pass through while blocking or reflecting others.**
- **Example:** A coffee filter that lets the liquid through but stops the grounds.

• Exercise 1: The Attenuator Problem

- An optical transmitter sends a signal at **+3 dBm**. The receiver has a maximum input power limit (saturation point) of **-5 dBm**. What is the minimum value of an **Optical Attenuator** needed to protect the receiver if the fiber loss is

Solution:

1. Current Power: +3 dBm
2. Target Power: -5 dBm
3. Required Reduction: $Power_{in} - Power_{max} = 3 \text{ dBm} - (-5 \text{ dBm}) = 8 \text{ dB}$.

- Answer: An attenuator of at least **8 dB** is required.

dBm ইউনিটে **প্লাস (+)** বা **মাইনাস (-)** চিহ্ন দিয়ে বোঝানো হয় যে সিগন্যালের শক্তি **১ মিলিওয়াট (1 mW)** থেকে বেশি নাকি কম। যেহেতু $0 \text{ dBm} = 1 \text{ mW}$, তাই এর ওপর ভিত্তি করে হিসাবটি করা হয়।

পাওয়ার (dBm)	পাওয়ার (mW) এ রূপান্তর	অবস্থা
+১০ dBm	১০ mW	অনেক শক্তিশালী সিগন্যাল
+৩ dBm	২ mW	শক্তিশালী সিগন্যাল
০ dBm	১ mW	রেফারেন্স পয়েন্ট
-৩ dBm	০.৫ mW	সিগন্যাল অর্ধেক হয়ে গেছে
-১০ dBm	০.১ mW	দুর্বল সিগন্যাল

$$P_{(mW)} = 10^{\frac{P_{(dBm)}}{10}}$$

• Exercise 2: Coupler Power Calculation

- A laser sends a signal with **10 mW** of power into a **1:4 symmetric coupler** (a coupler that splits the signal equally into 4 output fibers).
- Calculate the power in **mW** at each output port (assume no internal excess loss).
- Express the output power of one port in **dBm**.

Solution:

1. Power in mW:

Since it is a 1:4 symmetric split: $P_{out} = \frac{10 \text{ mW}}{4} = \mathbf{2.5 \text{ mW}}$.

2. Power in dBm:

Formula: $P_{(dBm)} = 10 \log_{10}(P_{mW})$

$$P_{(dBm)} = 10 \log_{10}(2.5)$$

$$P_{(dBm)} = 10 \times 0.3979 \approx \mathbf{3.98 \text{ dBm}}$$